CCC407: Chemistry in Our Life

Assignment # 1

Total marks: 10 × 2 = 20 August 28, 2026

Please ensure the assignment is submitted by **3 September, before 2:00 PM**.

1. Describe the chemistry behind the leavening process of baking powder.
2. Discuss the processes involved in protein denaturation under different conditions.
3. Explore the chemistry behind fireworks, focusing on how various chemicals create colors and effects.
4. Describe the importance of studying the composition of the atmosphere.
5. What chemical processes lead to the formation of acid rain? Explain how acid rain affects any copper material.
6. Which property of metals is primarily associated with the presence of mobile e^-?

   a. Brittleness

   b. Electrical conductivity

   c. Low density

   d. Low melting point

7. Which type of chemical bond exists in table salt?

   a. Ionic bond

   b. Covalent bond

   c. Metallic bond

   d. Hydrogen bond

8. Which property of water is responsible for its ability to act as a universal solvent?

    a. Low viscosity

    b. High thermal conductivity

    c. Polarity

    d. Anomalous expansion

9. Why does ice float on liquid water?

    a. Ice has a higher density than liquid water

    b. Water has maximum density at 4 °C

    c. Hydrogen bonding completely disappears in ice

    d. Ice contains more dissolved ions than water

10. What chemical process occurs when an antacid relieves stomach acidity?

    a. Combustion

    b. Polymerization

    c. Acid-base neutralization

    d. Fermentation